```
mysql> CREATE TABLE order_items (
    ->     order_id INT,
    ->     dish_price DECIMAL(10,2)
    -> );
Query OK, 0 rows affected (0.010 sec)

mysql> delimiter $$
mysql> create trigger update_order after insert on order_items for each row
    -> begin
    ->     update orders
    ->     set total_amount=total amount+new.dish_price
    ->     where order_id=new.order_id;
    -> end$$
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'amount+new.dish_price
    where order_id=new.order_id;
end' at line 4
mysql> CREATE TRIGGER update_order_total
    -> AFTER INSERT
    -> ON order_items
    -> FOR EACH ROW
    -> BEGIN
    ->     UPDATE orders
    ->     SET total_amount = total_amount + NEW.dish_price
    ->     WHERE order_id = NEW.order_id;
    -> END$$
Query OK, 0 rows affected (0.004 sec)
```

```
    ->         SIGNAL SQLSTATE '45000'
    ->         SET MESSAGE_TEXT = 'Salary cannot be negative';
    ->     END IF;
    -> END$$
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'OR UPDATE
ON employees
FOR EACH ROW
BEGIN
    IF NEW.salary < 0 THEN
      SIG' at line 2
mysql>
mysql> DELIMITER ;
mysql> DELIMITER $$
mysql>
mysql> CREATE TRIGGER prevent_negative_salary_insert
    -> BEFORE INSERT ON employees
    -> FOR EACH ROW
    -> BEGIN
    ->     IF NEW.salary < 0 THEN
    ->         SIGNAL SQLSTATE '45000'
    ->         SET MESSAGE_TEXT = 'Salary cannot be negative';
    ->     END IF;
    -> END$$
Query OK, 0 rows affected (0.004 sec)

mysql>
mysql> DELIMITER ;
mysql> DELIMITER $$
mysql>
mysql> CREATE TRIGGER prevent_negative_salary_update
    -> BEFORE UPDATE ON employees
    -> FOR EACH ROW
    -> BEGIN
    ->     IF NEW.salary < 0 THEN
    ->         SIGNAL SQLSTATE '45000'
    ->         SET MESSAGE_TEXT = 'Salary cannot be negative';
    ->     END IF;
    -> END$$
Query OK, 0 rows affected (0.006 sec)

mysql>
mysql> DELIMITER ;
mysql> DELIMITER $$
mysql>
mysql> CREATE TRIGGER employee_update_audit
    -> AFTER UPDATE ON employees
    -> FOR EACH ROW
    -> BEGIN
    ->     INSERT INTO employee_audit (
    ->         emp_id,
    ->         old_name,
    ->         updated_at
    ->     )
    ->     VALUES (
    ->         OLD.emp_id,
    ->         OLD.name,
    ->         CURRENT_TIMESTAMP
    ->     );
    -> END$$
Query OK, 0 rows affected (0.008 sec)

mysql>
mysql> DELIMITER ;
mysql>
```

```
mysql> use online_banking_system;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> select DATEDIFF('2026-02-04','2026-02-02') as no_of_days from users;
+------------+
| no_of_days |
+------------+
|          2 |
|          2 |
|          2 |
|          2 |
+------------+
4 rows in set (0.005 sec)

mysql> select DATE_ADD('2026-02-01',INTERVAL 30 DAY) AS expiry_date from users;
+-------------+
| expiry_date |
+-------------+
| 2026-03-03  |
| 2026-03-03  |
| 2026-03-03  |
| 2026-03-03  |
+-------------+
4 rows in set (0.001 sec)
```

```
mysql> select DATE_SUB(curdate(),INTERVAL 7 DAY) AS BEFORE_7_DAYS from users;
+---------------+
| BEFORE_7_DAYS |
+---------------+
| 2026-01-28    |
| 2026-01-28    |
| 2026-01-28    |
| 2026-01-28    |
+---------------+
4 rows in set (0.001 sec)

mysql> select * from users where daoofweek(created_date) in (1,7);
ERROR 1305 (42000): FUNCTION online_banking_system.daoofweek does not exist
mysql> select * from users where dayofweek(created_date) in (1,7);
ERROR 1054 (42S22): Unknown column 'created_date' in 'where clause'
mysql> select * from users where DAYOFWEEK(created_date) in (1,7);
ERROR 1054 (42S22): Unknown column 'created_date' in 'where clause'
mysql> select * from transactions where dayofweek(created_date) in (1,7);
ERROR 1054 (42S22): Unknown column 'created_date' in 'where clause'
mysql> describe transactions;
+------------------+---------------------------------------+------+-----+-------------------+-----------------+
| Field            | Type                                  | Null | Key | Default           | Extra           |
+------------------+---------------------------------------+------+-----+-------------------+-----------------+
| transaction_id   | bigint                                | NO   | PRI | NULL              | auto_increment  |
| from_account     | bigint                                | YES  | MUL | NULL              |                 |
| to_account       | bigint                                | YES  | MUL | NULL              |                 |
| amount           | decimal(15,2)                         | NO   |     | NULL              |                 |
```

```
| transaction_type | enum('deposit','withdrawal','transfer') | NO   |     | NULL            |
|

| status         | enum('success','failed')             | NO   |     | NULL            |               |

| created_at      | timestamp                            | YES  |     | CURRENT_TIMESTAMP |
DEFAULT_GENERATED |

+------------------+----------------------------------------+------+-----+-------------------+------------
-------+

7 rows in set (0.007 sec)


mysql> SELECT *

   -> FROM transactions

   -> WHERE DAYOFWEEK(created_at) IN (1, 7);

Empty set (0.003 sec)


mysql> select * from transactions where DAYOFWEEK(created_at) in (3,7);

Empty set (0.001 sec)


mysql> select MONTHNAME(str_to_date('2026-02-04','%y-%m-%d'))as month_name;

+------------+

| month_name |

+------------+

| NULL       |

+------------+

1 row in set, 1 warning (0.002 sec)


mysql> select MONTHNAME(str_to_date('2026-02-04','%Y-%m-%d'))as month_name;

+------------+

| month_name |

+------------+
```

```
| February   |
+------------+
1 row in set (0.001 sec)

mysql> describe users;
+------------+-------------------------+------+-----+-------------------+-------------------+
| Field      | Type                    | Null | Key | Default           | Extra             |
+------------+-------------------------+------+-----+-------------------+-------------------+
| USER_ID    | bigint                  | NO   | PRI | NULL              | auto_increment    |
| NAME       | varchar(89)             | NO   |     | NULL              |                   |
| EMAIL      | varchar(89)             | YES  |     | NULL              |                   |
| PASSWORD   | varchar(789)            | NO   |     | NULL              |                   |
| ROLE       | enum('CUSTOMER','ADMIN') | YES  |     | CUSTOMER          |                   |
| Created_at | timestamp               | YES  |     | CURRENT_TIMESTAMP |
DEFAULT_GENERATED |
+------------+-------------------------+------+-----+-------------------+-------------------+
6 rows in set (0.002 sec)

mysql> create INDEX IDS_EMAIL on users(email);
Query OK, 0 rows affected (0.033 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> describe users;
+------------+-------------------------+------+-----+-------------------+-------------------+
| Field      | Type                    | Null | Key | Default           | Extra             |
+------------+-------------------------+------+-----+-------------------+-------------------+
| USER_ID    | bigint                  | NO   | PRI | NULL              | auto_increment    |
| NAME       | varchar(89)             | NO   |     | NULL              |                   |
```

| EMAIL | varchar(89) | YES | MUL | NULL | |

| PASSWORD | varchar(789) | NO | | NULL | |

| ROLE | enum('CUSTOMER','ADMIN') | YES | | CUSTOMER | |

| Created_at | timestamp | YES | | CURRENT_TIMESTAMP | DEFAULT_GENERATED |

+----------+----------+------+-----+----------+----------+

6 rows in set (0.002 sec)

mysql> select 8from users;

ERROR 1054 (42S22): Unknown column '8from' in 'field list'

mysql> select *fro useers;

ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'fro useers' at line 1

mysql> select *from users;

+---------+----------+----------+----------+----------+----------+

| USER_ID | NAME | EMAIL | PASSWORD | ROLE | Created_at |

+---------+----------+----------+----------+----------+----------+

| 1 | MANOJ | kotkurumanojkumar@gmail.com | 1235678 | CUSTOMER | 2026-01-02 10:55:01 |

| 2 | AKASH | akash@gmail.com | akash123 | ADMIN | 2026-01-02 10:55:01 |

| 3 | JITENDRA | jithendra@gmail.com | jithu123 | CUSTOMER | 2026-01-02 10:55:01 |

| 4 | JASWANTH | jaswanth@gmail.com | jashu123 | ADMIN | 2026-01-02 10:55:01 |

+---------+----------+----------+----------+----------+----------+

4 rows in set (0.001 sec)

mysql> select *from users where email='kotkurumanojkumar@gmail.com';

+---------+-------+----------+----------+----------+----------+

```
| USER_ID | NAME  | EMAIL                    | PASSWORD | ROLE     | Created_at        |
+---------+-------+-----------------------------+----------+----------+---------------------+
|       1 | MANOJ | kotkurumanojkumar@gmail.com | 1235678  | CUSTOMER | 2026-01-02 10:55:01 |
+---------+-------+-----------------------------+----------+----------+---------------------+
1 row in set (0.001 sec)
```

```
mysql> mysql> create unique index idx_username on users(name);
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'name)' at line 1
mysql> create unique INDEX idx_username on users('name');
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near ''name')' at line 1
mysql> create unique INDEX idx_username on users('NAME');
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near ''NAME')' at line 1
mysql> create UNIQUE INDEX idx_username on users('NAME');
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near ''NAME')' at line 1
mysql> alter table users change NAME username varchar(100);
Query OK, 0 rows affected (0.042 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> create UNIQUE INDEX idx_username on users(username);
Query OK, 0 rows affected (0.024 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

```
mysql> create unique INDEX IDX_EMAIL on users(email);

Query OK, 0 rows affected (0.034 sec)

Records: 0  Duplicates: 0  Warnings: 0


mysql> select*from users;

+---------+----------+----------------------------+----------+----------+---------------------+

| USER_ID | username | EMAIL                      | PASSWORD | ROLE     | Created_at          |

+---------+----------+----------------------------+----------+----------+---------------------+

|       1 | MANOJ    | kotkurumanojkumar@gmail.com | 1235678  | CUSTOMER | 2026-01-02 10:55:01 |

|       2 | AKASH    | akash@gmail.com            | akash123 | ADMIN    | 2026-01-02 10:55:01 |

|       3 | JITENDRA | jithendra@gmail.com        | jithu123 | CUSTOMER | 2026-01-02 10:55:01 |

|       4 | JASWANTH | jaswanth@gmail.com         | jashu123 | ADMIN    | 2026-01-02 10:55:01 |

+---------+----------+----------------------------+----------+----------+---------------------+

4 rows in set (0.001 sec)


mysql> select index fromusers;

ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'index fromusers' at line 1

mysql> select index from users;

ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'index from users' at line 1

mysql> select INDEX from users;

ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'INDEX from users' at line 1

mysql> SHOW INDEX from users;
```

```
+-------+------------+--------------+--------------+-------------+-----------+-------------+----------+--
------+------+------------+---------+---------------+---------+------------+

| Table | Non_unique | Key_name     | Seq_in_index | Column_name | Collation |
Cardinality | Sub_part | Packed | Null | Index_type | Comment | Index_comment | Visible |
Expression |

+-------+------------+--------------+--------------+-------------+-----------+-------------+----------+--
------+------+------------+---------+---------------+---------+------------+

| users |          0 | PRIMARY      |            1 | USER_ID     | A         |           4 |     NULL |   NULL |      |
BTREE      |         |             | YES     | NULL       |

| users |          0 | idx_username |            1 | username    | A         |           4 |     NULL |   NULL | YES
| BTREE      |         |             | YES     | NULL       |

| users |          0 | IDX_EMAIL    |            1 | EMAIL       | A         |           4 |     NULL |   NULL | YES  |
BTREE      |         |             | YES     | NULL       |

| users |          1 | IDS_EMAIL    |            1 | EMAIL       | A         |           4 |     NULL |   NULL | YES  |
BTREE      |         |             | YES     | NULL       |

+-------+------------+--------------+--------------+-------------+-----------+-------------+----------+--
------+------+------------+---------+---------------+---------+------------+

4 rows in set (0.007 sec)


mysql> mysql> drop idx_email(akash@email.com) on users;

ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that
corresponds to your MySQL server version for the right syntax to use near
'(akash@email.com) on users' at line 1

mysql> drop index iidx_email on users;

ERROR 1091 (42000): Can't DROP 'iidx_email'; check that column/key exists

mysql> drop INDEX idx_email on users;

Query OK, 0 rows affected (0.018 sec)

Records: 0  Duplicates: 0  Warnings: 0


mysql> drop INDEX idx_email(akash@gmail.com) on users;
```

ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near '(akash@gmail.com) on users' at line 1

mysql> show index on users;

ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'on users' at line 1

mysql> show index from users;

```
+-------+------------+--------------+--------------+-------------+-----------+-------------+----------+--
------+------+------------+---------+---------------+---------+------------+
| Table | Non_unique | Key_name     | Seq_in_index | Column_name | Collation |
Cardinality | Sub_part | Packed | Null | Index_type | Comment | Index_comment | Visible |
Expression |
+-------+------------+--------------+--------------+-------------+-----------+-------------+----------+--
------+------+------------+---------+---------------+---------+------------+
| users |          0 | PRIMARY      |            1 | USER_ID     | A         |           4 |     NULL |   NULL |     |
BTREE     |         |            | YES     | NULL       |
| users |          0 | idx_username |            1 | username    | A         |           4 |     NULL |   NULL | YES
| BTREE     |         |            | YES     | NULL       |
| users |          1 | IDS_EMAIL    |            1 | EMAIL       | A         |           4 |     NULL |   NULL | YES  |
BTREE     |         |            | YES     | NULL       |
+-------+------------+--------------+--------------+-------------+-----------+-------------+----------+--
------+------+------------+---------+---------------+---------+------------+
3 rows in set (0.003 sec)
```

mysql> create INDEX idx_aadhar on users(aadhar);

ERROR 1072 (42000): Key column 'aadhar' doesn't exist in table

mysql> describe accounts;

```
+----------------+--------------------------+------+-----+-------------------+-------------------+
| Field          | Type                     | Null | Key | Default           | Extra             |
+----------------+--------------------------+------+-----+-------------------+-------------------+
| account_id     | bigint                   | NO   | PRI | NULL              | auto_increment    |
```

```
| user_id       | bigint                | NO   | MUL | NULL            |                |
| account_number | varchar(20)            | NO   | UNI | NULL            |                |
| account_type   | enum('savings','current') | NO   |     | NULL            |                |
| balance       | decimal(15,2)          | NO   |     | 0.00            |                |
| status        | enum('active','blocked')  | YES  |     | active          |                |
| created_at    | timestamp              | YES  |     | CURRENT_TIMESTAMP |
DEFAULT_GENERATED |
+----------------+---------------------------+------+-----+-------------------+-------------------+
7 rows in set (0.004 sec)

mysql> select*from accounts;
+------------+---------+----------------+--------------+-----------+--------+---------------------+
| account_id | user_id | account_number | account_type | balance   | status | created_at
|
+------------+---------+----------------+--------------+-----------+--------+---------------------+
|          5 |       1 | SBI0000001     | savings      |  75000.00 | active | 2026-01-02 11:00:23 |
|          6 |       2 | SBI0000002     | savings      | 120000.50 | active | 2026-01-02 11:00:23 |
|          7 |       3 | SBI0000003     | savings      |  50000.75 | active | 2026-01-02 11:00:23 |
|          8 |       4 | SBI0000004     | savings      |  95000.00 | active | 2026-01-02 11:00:23 |
+------------+---------+----------------+--------------+-----------+--------+---------------------+
4 rows in set (0.003 sec)

mysql> create INDEX idx_account_number on users(account_number);
ERROR 1072 (42000): Key column 'account_number' doesn't exist in table
mysql> create INDEX idx_account_number on accounts(account_number);
Query OK, 0 rows affected (0.027 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> select*from accounts where account_number='SBI0000001';
```

```
+------------+---------+----------------+--------------+----------+--------+---------------------+
| account_id | user_id | account_number | account_type | balance  | status | created_at
|
+------------+---------+----------------+--------------+----------+--------+---------------------+
|          5 |       1 | SBI0000001     | savings      | 75000.00 | active | 2026-01-02 11:00:23 |
+------------+---------+----------------+--------------+----------+--------+---------------------+
1 row in set (0.001 sec)

mysql> create INDEX idx_s_cr on users(status,created_at);
ERROR 1072 (42000): Key column 'status' doesn't exist in table
mysql> create INDEX idx_s_cr on accounts(status,created_at);
Query OK, 0 rows affected (0.029 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> select*from accounts where status='active' order by created_at;
+------------+---------+----------------+--------------+-----------+--------+---------------------+
| account_id | user_id | account_number | account_type | balance   | status | created_at
|
+------------+---------+----------------+--------------+-----------+--------+---------------------+
|          5 |       1 | SBI0000001     | savings      |  75000.00 | active | 2026-01-02 11:00:23 |
|          6 |       2 | SBI0000002     | savings      | 120000.50 | active | 2026-01-02 11:00:23 |
|          7 |       3 | SBI0000003     | savings      |  50000.75 | active | 2026-01-02 11:00:23 |
|          8 |       4 | SBI0000004     | savings      |  95000.00 | active | 2026-01-02 11:00:23 |
+------------+---------+----------------+--------------+-----------+--------+---------------------+
4 rows in set (0.001 sec)

mysql> select email,concat(substring(email,'@',1),'123') as password from users;
+----------------------------+----------+
| email                      | password |
```

```
+-----------------------------+----------+
| akash@gmail.com           | 123     |
| jaswanth@gmail.com        | 123     |
| jithendra@gmail.com       | 123     |
| kotkurumanojkumar@gmail.com | 123     |
+-----------------------------+----------+
4 rows in set, 5 warnings (0.003 sec)

mysql> describe veltech;
+------------+-------------+------+-----+---------+-------+
| Field     | Type       | Null | Key | Default | Extra |
+------------+-------------+------+-----+---------+-------+
| user_id    | int        | YES |    | NULL    |      |
| name       | varchar(45) | YES |    | NULL    |      |
| account_no | varchar(78) | YES |    | NULL    |      |
| acc_type   | varchar(34) | YES |    | NULL    |      |
| balance    | int        | YES |    | NULL    |      |
+------------+-------------+------+-----+---------+-------+
5 rows in set (0.003 sec)

mysql> describe customers;;
+---------------+-------------+------+-----+---------+-------+
| Field        | Type       | Null | Key | Default | Extra |
+---------------+-------------+------+-----+---------+-------+
| customer_id   | int        | YES |    | NULL    |      |
| name         | varchar(89) | YES |    | NULL    |      |
| email        | varchar(67) | YES |    | NULL    |      |
| phone_no     | varchar(89) | YES |    | NULL    |      |
```

```
| date_of_birth | date       | YES |    | NULL    |      |
| status        | varchar(45) | YES |    | NULL    |      |
+---------------+-------------+------+-----+---------+-------+
6 rows in set (0.003 sec)

ERROR:
No query specified

mysql> select *from customers;
+-------------+-------+-----------------------+------------+---------------+--------+
| customer_id | name  | email                 | phone_no   | date_of_birth | status |
+-------------+-------+-----------------------+------------+---------------+--------+
|       25025 | MANOJ | kotkurumanojkumar     | 8317607955 | 2025-01-27    | active |
|       26567 | MANOJ | manoj@gmail.com       | 9010030003 | 2025-01-27    | active |
|       25024 | akash | bandaruakash@gmail.com | 9014744204 | 2025-01-27    | active |
+-------------+-------+-----------------------+------------+---------------+--------+
3 rows in set (0.003 sec)

mysql> select contact, reg_replace(contact,'[^0-9]','') as valid contact from customers;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'contact from customers' at line 1
mysql> select name,concat(upper(substring(name,1,1)),lower(substring(name,2))) as normal_name from customers;
+-------+-------------+
| name  | normal_name |
+-------+-------------+
| MANOJ | Manoj       |
| MANOJ | Manoj       |
```

```
| akash | Akash      |
+-------+-------------+
3 rows in set (0.001 sec)
```

```
mysql> select contact, REGEXP_REPLACE(contact,'[^0-9]','') as valid contact from customers;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'contact from customers' at line 1
mysql> select contact, REGEXP_REPLACE(contact,'[^0-9]','') as valid_contact from customers;
ERROR 1054 (42S22): Unknown column 'contact' in 'field list'
mysql> select phone_no, REGEXP_REPLACE(phone_no,'[^0-9]','') as valid_contact from customers;
+------------+---------------+
| phone_no   | valid_contact |
+------------+---------------+
| 8317607955 | 8317607955    |
| 9010030003 | 9010030003    |
| 9014744204 | 9014744204    |
+------------+---------------+
3 rows in set (0.016 sec)
```

```
mysql> select name, coalesce(customer_id,phone_no)as preffered_contact from customers;
+-------+-------------------+
| name  | preffered_contact |
+-------+-------------------+
| MANOJ | 25025             |
| MANOJ | 26567             |
```

```
| akash | 25024          |
+-------+-------------------+
3 rows in set (0.001 sec)

mysql> select customer_id, avg(phone_no) as avg_phone_no from customers group by
customer_id;
+-------------+--------------+
| customer_id | avg_phone_no |
+-------------+--------------+
|      25025 |   8317607955 |
|      26567 |   9010030003 |
|      25024 |   9014744204 |
+-------------+--------------+
3 rows in set (0.003 sec)

mysql> select customer_id, avg(phone_no) as avg_phone_no from customers;
ERROR 1140 (42000): In aggregated query without GROUP BY, expression #1 of SELECT
list contains nonaggregated column 'online_banking_system.customers.customer_id';
this is incompatible with sql_mode=only_full_group_by
mysql> SELECT AVG(phone_no) AS avg_phone_no
    -> FROM customers;
+--------------+
| avg_phone_no |
+--------------+
|   8780794054 |
+--------------+
1 row in set (0.001 sec)

mysql>
```